

NMDPRA

AI DATA INTELLIGENCE DASHBOARD

A Proposal for AI-Powered Petroleum Data Analysis

Submitted To
**Nigerian Midstream and Downstream
Petroleum Regulatory Authority**
ICT Department

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Executive Summary

During my service year with NMDPRA's ICT Support Department, I identified a critical gap: petroleum truckout data exists in abundance, but extracting meaningful insights from it requires technical skills most staff do not have. Analysts build reports manually. Decision makers wait days for answers. Non-technical staff cannot access the data at all.

I built a solution. The NMDPRA AI Data Intelligence Dashboard allows any staff member to type a question in plain English and receive an instant chart, map, or table with a plain English explanation. No SQL. No Excel. No technical training required.

This document presents the prototype, its capabilities, its impact on NMDPRA operations, and a roadmap for what it becomes in Version 2.

01 | The Problem NMDPRA Currently Faces

NMDPRA generates and holds significant petroleum truckout data — records covering every product movement across Nigeria's states, depots, transporters, refineries, and customers. This data has enormous potential to inform regulatory decisions, identify supply gaps, monitor compliance, and support strategic planning.

However, the current reality is that this potential largely goes unrealised because of one fundamental barrier:

The people who need insights from the data are not the people who can extract them.

Managers and directors who make decisions cannot query databases. Non-technical staff cannot write Excel formulas for complex aggregations. Even when analysts produce reports, the process is manual, time-consuming, and produces static outputs that cannot be interrogated further.

Specific Pain Points

Current Situation	The Cost
Reports are built manually by analysts	Time lost. Human error risk. Analyst bottleneck.
Non-technical staff cannot access data independently	Over-reliance on a small number of people.
Static reports cannot be questioned or explored	Decision makers get answers but cannot dig deeper.
Comparing products, states, or time periods requires new reports	Slow response to emerging supply issues.
Data visualisation requires specialist tools and training	Power BI skills are not universally available.

02 | The Solution — NMDPRA AI Data Intelligence Dashboard

The NMDPRA AI Data Intelligence Dashboard is a working prototype that transforms how petroleum truckout data is accessed and understood. It eliminates the technical barrier entirely by replacing queries, formulas, and report requests with a single natural language interface.

Any staff member types a question in plain English.
The system interprets the question using artificial intelligence.
A chart, map, or table is generated instantly from the actual data.
A plain English explanation is written automatically describing what the result means.

The dashboard was built specifically for NMDPRA using your actual data structure — the exact column names, product types, Nigerian states, depot names, and supply categories that appear in your petroleum truckout records. This is not a generic tool. It was designed for this organisation and this data.

Example Questions the System Can Answer

- *Which states received the most PMS in January 2026?*
- *Show me a map of total quantity loaded per destination state.*
- *Which transporters moved the highest volume of AGO?*
- *What is the breakdown of product supply types as a pie chart?*
- *Show the daily trend of DPK quantity loaded over time.*
- *PMS quantity loaded from Dangote Refinery delivered to Lagos.*
- *Compare bridging vs local distribution by product.*
- *What are the top 10 customers by quantity received?*
- *Show me the quantity loaded in Kano from 12th January 2026 to 30th January 2026.*

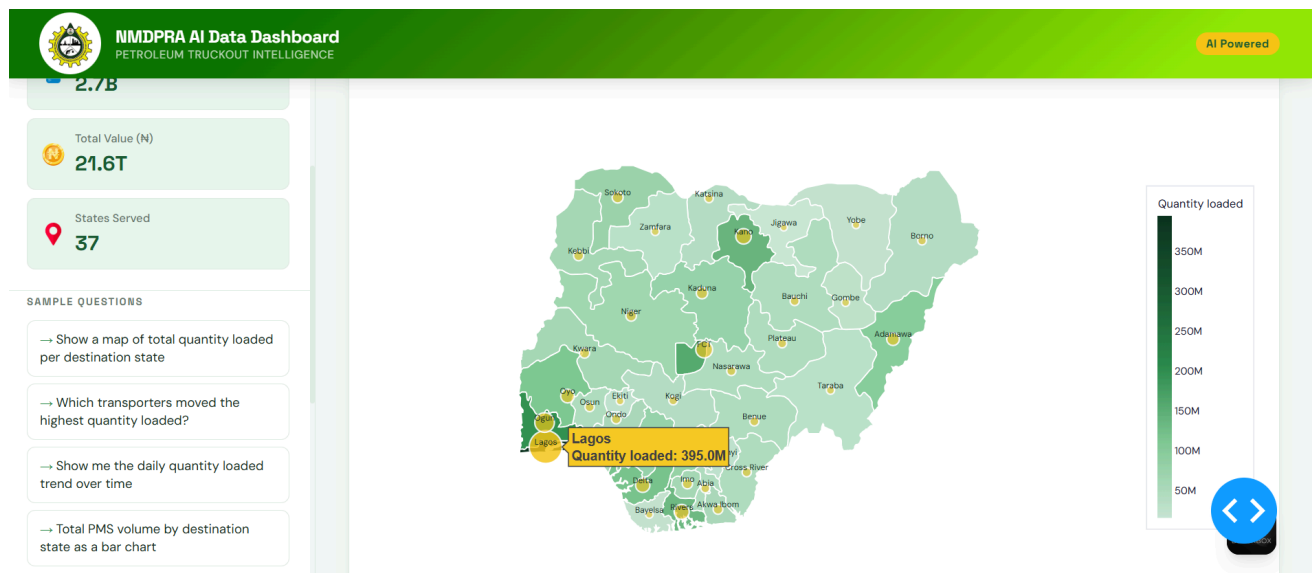
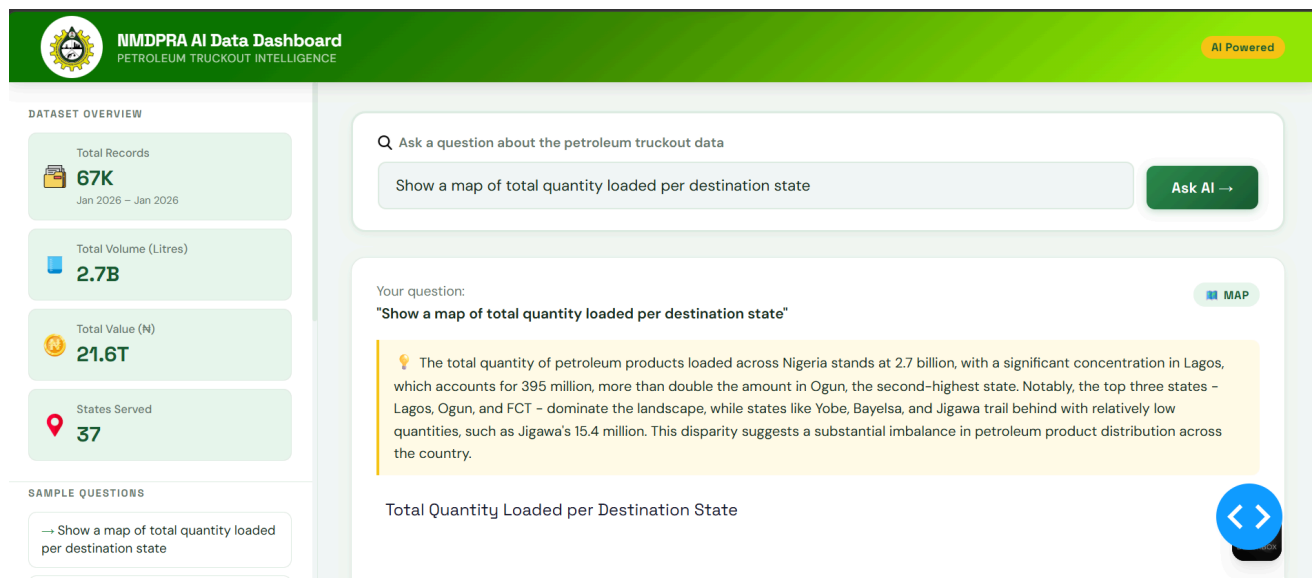


Figure 1 & 2: NMDPRA AI Data Intelligence Dashboard — Nigeria petroleum distribution map generated from a plain English question in under 2 seconds.

03 | Current Features (Version 1)

#	Feature	Description
1	Natural Language Queries	Staff type questions in plain English. No technical knowledge, SQL, or Excel required.
2	Automatic Chart Generation	Bar charts, line charts, pie charts, and scatter plots generated automatically based on the question.
3	Nigeria Choropleth Map	State-level distribution maps using official Nigeria GeoJSON boundaries. States are colour-filled by value with proportional bubble overlays.
4	AI-Generated Explanations	Every result includes a plain English insight written by the AI describing what the data actually means — not just what it shows.
5	Multi-Filter Support	Questions can filter by up to three criteria simultaneously — e.g. product, refinery, and destination state at once.
6	Date Range Filtering	Natural date phrases like 'first two weeks of January' or 'last 30 days' are automatically converted to exact date filters.
7	Sample Questions	A sidebar of pre-written questions guides users who are unsure where to start.
8	Query History	Recent questions are tracked in the sidebar so users can reference previous queries.
9	Graceful Error Handling	Off-topic or unclear questions receive a friendly explanation and suggested alternatives rather than crashing.
10	NMDPRA Branding	Full NMDPRA visual identity — official logo, green and yellow colour palette, professional typography.

04 | How It Works

The system is built on a simple but powerful principle: the artificial intelligence does not touch your data. It only translates your question into instructions. Everything else happens locally on the organisation's machine.

Step 1: Staff member types a question in plain English and clicks Ask AI.

Step 2: The question is sent to an AI language model along with a description of the dataset structure — column names and example values only. No actual data is transmitted.

Step 3: The AI interprets the question and returns structured instructions: chart type, which columns to use, what filters to apply, what date range to use.

Step 4: Your data is queried locally on your machine using Python. The data never leaves the organisation's infrastructure.

Step 5: A chart, map, or table is built from the query results using Plotly — a professional data visualisation library.

Step 6: A second AI call receives the actual data results and writes a plain English explanation — specific numbers, trends, and insights.

Step 7: The chart and explanation appear on screen. The entire process takes approximately 2 seconds.

DATA SECURITY NOTE:

The actual petroleum records, volumes, prices, customer names, and truck numbers never leave the organisation's machine.

The only information transmitted to the AI is: the user's question + a list of column names and example values.

All data processing, filtering, and aggregation happens locally using Python.

05 | Positive Impact on NMDPRA Operations

The following impacts are specific, measurable, and directly tied to NMDPRA's mandate as Nigeria's petroleum midstream and downstream regulatory authority.

#	Impact Area	Specific Benefit to NMDPRA
01	Faster Regulatory Decisions	Directors and managers can interrogate petroleum distribution data in real time without waiting for analyst-produced reports. A question that previously took a day to answer now takes 2 seconds.
02	Reduced Analyst Workload	Manual report building is eliminated for standard queries. Analysts are freed from repetitive data extraction tasks and can focus on deeper analysis and strategic work.
03	Wider Data Access Across Staff	Non-technical staff in any department can access and explore petroleum data independently. Data literacy becomes organisational rather than individual.
04	Supply Gap Identification	The map visualisation makes it immediately visible which states are receiving low product volumes. Regulatory attention can be directed quickly to underserved regions.
05	Transporter and Depot Monitoring	Instant ranking of transporters and source depots by volume makes performance monitoring and compliance oversight significantly more efficient.
06	Product Distribution Oversight	The ability to filter by product type, refinery, destination state, and date simultaneously gives regulators granular visibility into how specific products move through the supply chain.
07	Evidence-Based Reporting	When NMDPRA prepares regulatory reports or stakeholder briefings, staff can generate supporting charts and maps in minutes rather than hours.
08	Cost Efficiency	The system runs at near-zero cost on existing hardware. There are no additional software licenses, no Power BI Premium fees, no data consultant costs for standard queries.
09	Technology Leadership	NMDPRA would be among the first Nigerian government regulatory agencies to deploy a conversational AI data intelligence tool. This positions the authority as a forward-thinking, technology-enabled regulator.
10	Foundation for Advanced Analytics	Version 1 establishes the data pipeline, AI integration, and visualisation infrastructure. Version 2 and beyond can add predictive analytics, anomaly detection, and automated regulatory alerts at relatively low incremental cost.

06 | Comparison With Existing Tools

NMDPRA currently uses Microsoft Power BI and Excel for data analysis. The AI Dashboard does not replace these tools — it complements them. However, in specific use cases it offers capabilities neither tool can match.

Capability	Power BI / Excel	AI Dashboard
Plain English question interface	No	Yes — core feature
Usable by non-technical staff immediately	Requires training	Yes — no training needed
AI-generated insight explanation	No	Yes — every result
Nigeria state choropleth map	Requires configuration	Built-in, ready to use
Multi-filter natural language queries	Requires manual filter setup	Yes — automatic
Relative date filtering (e.g. last 30 days)	Manual date picker	Plain English phrases
Cost for internal use	Per-user licensing fees	Near zero
Custom NMDPRA branding	Requires Pro/Premium	Built-in
Drag-and-drop report building	Yes	Not applicable
Complex multi-page dashboards	Yes	Version 2 roadmap

07 | Technical Foundation

The dashboard is built entirely on open-source, production-grade technologies used by leading data organisations worldwide. Every technology choice was made intentionally for reliability, performance, and long-term maintainability.

Technology	Component	Why It Was Chosen
Python	Core language	Industry standard for data science. Extensive library ecosystem. Free and open source.
Dash by Plotly	Web framework	Python-native web application framework built specifically for data dashboards.
Plotly	Visualisation	Professional interactive charts and maps. Used by Fortune 500 companies and government agencies.
Pandas	Data processing	The leading Python library for data manipulation. Handles millions of rows efficiently.
Parquet	Data format	Compressed, high-performance data format. Loads significantly faster than CSV.
Groq API	AI inference	Ultra-fast LLM inference platform. Free tier supports approximately 700 queries per day.
LLaMA 3.3 70B	Language model	Meta's open-source language model. Excellent at structured data analysis tasks.
Nigeria GeoJSON	Map data	Official state boundary data for accurate Nigeria state map rendering.

08 | Version 2 Roadmap

Version 1 is a functional prototype demonstrating the core concept. Version 2 transforms it into a production-ready organisational tool. The following features are prioritised based on operational impact.

High Priority

- User authentication and login page — controlled access by staff role
- Role-based permissions — some users view only, admins manage data
- PNG and CSV export — download any chart or table directly
- Live data connection — dashboard always shows current records without manual file replacement
- Conversation memory — follow-up questions that remember the previous result

Medium Priority

- Team deployment — accessible from any computer on the NMDPRA network
- Multi-chart dashboard view — show 4-6 KPI charts simultaneously
- Scheduled email reports — automatic weekly summary delivered to management
- Mobile responsive layout — works on phones and tablets

Future Enhancements

- Anomaly detection — automatic alerts when unusual patterns are detected in supply data
- Predictive analytics — forecast future volumes based on historical trends
- PDF report builder — generate formatted regulatory reports from selected charts
- Audit log — track all queries by user for governance and accountability
- Multiple dataset support — switch between different data periods or sources

Infrastructure

- Production server deployment with HTTPS encryption
- Automated daily database backups
- Uptime monitoring and error alerting
- Performance optimisation for larger datasets

09 | Cost Analysis

One of the most significant advantages of this system is its exceptionally low cost relative to its capability. The following represents the current and projected cost structure.

Version 1 — Current Cost

Item	Monthly Cost	Notes
Groq AI API (LLaMA 3.3 70B)	NGN 0	Free tier — ~700 queries/day included
Python and all libraries	NGN 0	Open source — no licensing fees
Server / hosting	NGN 0	Runs on existing NMDPRA hardware
Maintenance (Version 1)	NGN 0	No external dependencies to maintain
TOTAL	NGN 0	Zero recurring cost for prototype

For context: Microsoft Power BI Pro costs approximately \$10 USD per user per month. For 10 users that is \$100/month — approximately NGN 160,000/month at current exchange rates.

This AI Dashboard delivers comparable and in some areas superior capability at zero cost.

10 | Recommendation and Next Steps

This prototype demonstrates that AI-powered data intelligence is not only possible for NMDPRA but is already working on your data today. The question is not whether to adopt it — the question is how quickly to develop it into a full organisational tool.

Recommended Immediate Actions

1. Formal Review: Schedule a demonstration for ICT leadership and relevant department heads. Allow staff to ask their own questions using the live system.

2. Pilot Programme: Deploy Version 1 to a small group of 2-5 staff members for a 30-day pilot. Collect feedback on which features add most value and which questions come up most often.

3. Version 2 Development: Based on pilot feedback, commission Version 2 development with the highest-priority features: user authentication, live data connection, and export functionality.

4. Organisational Deployment: Deploy the production version across relevant departments. Conduct brief staff orientation sessions — the tool requires minimal training by design.

This system was designed, built, and is being proposed by a NYSC corps member who served in the ICT Support Department.

It was built specifically for NMDPRA using your actual data structure, your branding, and an understanding of your operational context gained during the service year.

The expertise to develop, maintain, and expand this system is available. The infrastructure is already here. The data is ready.

The only remaining step is the decision to move forward.